

AWS Hands-On Project Report

Hosting a Static Website on Amazon S3

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Project Type: AWS Hands-On Practical
AWS Service: Amazon S3
Region: ap-south-1 (Mumbai)
Status: Completed

1. Project Overview

Project Title: Hosting a Static Website on Amazon S3

Objective

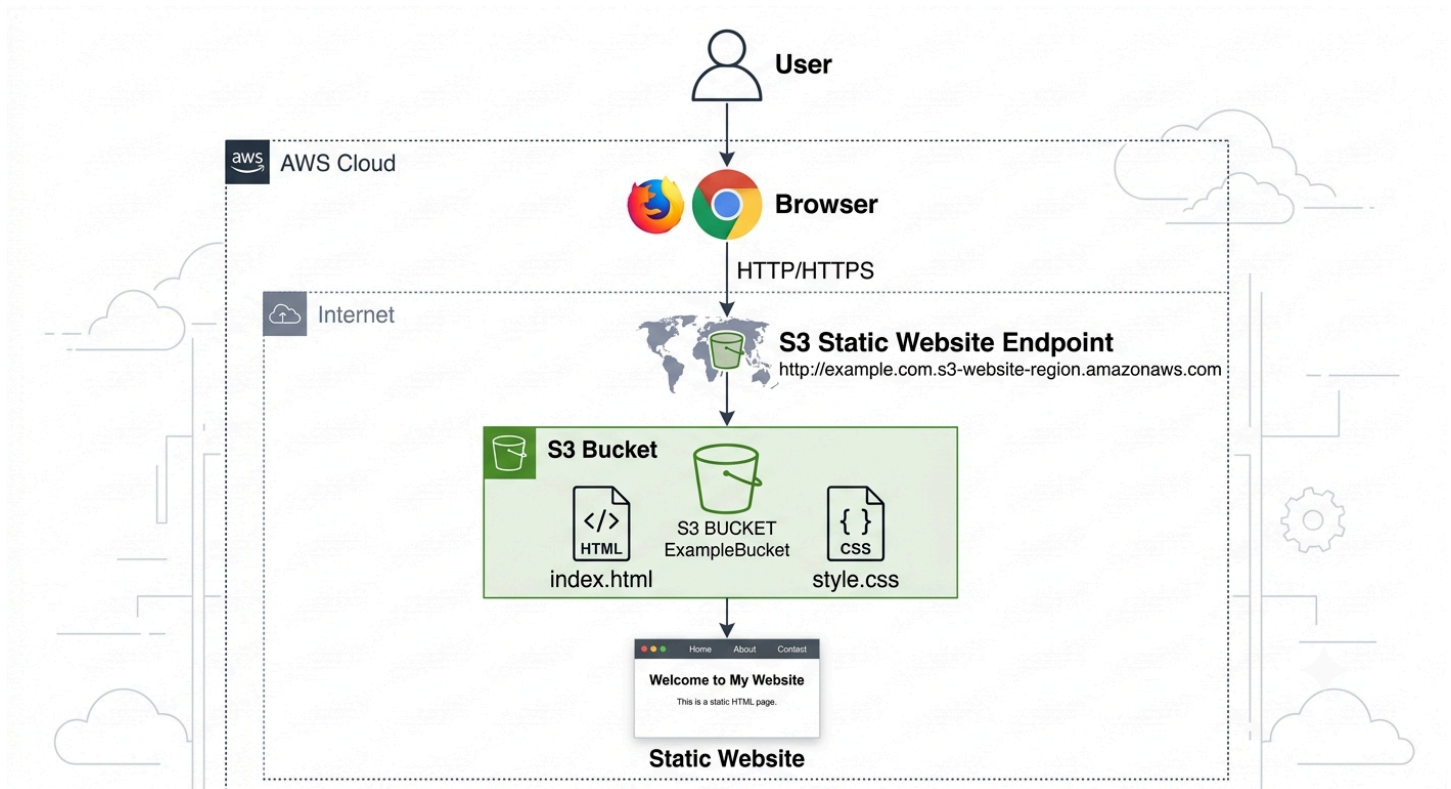
The objective of this project was to understand how Amazon S3 can be used to host a static website without deploying or managing a traditional web server such as Apache or Nginx.

In this practical, a simple HTML/CSS website was uploaded to an S3 bucket, static website hosting was enabled, the required access permissions were configured, and the website was accessed through the S3 website endpoint.

2. AWS Services and Technologies Used

Technology / Service	Purpose
Amazon S3	Store and host static website files
S3 Static Website Hosting	Serve the website through an S3 website endpoint
S3 Bucket Policy	Allow public read access to website objects
HTML	Website structure and content
CSS	Website styling
Web Browser	Test and access the hosted website

3. Project Architecture



Architecture Explanation

1. The user opens the S3 website endpoint in a browser.
2. The request reaches the S3 bucket through the website endpoint.
3. S3 retrieves the configured index document.
4. `index.html` references `style.css`.
5. S3 serves the required objects.
6. The browser renders the static website.

4. Project Files

The website used in this practical contains:

s3-static-website/

```
|
|
|—— index.html
|
|—— style.css
```

`index.html`: The HTML file contains the structure and content of the website.

`style.css`: The CSS file provides styling and presentation for the webpage.

5. Step 1 — Create an S3 Bucket

I created a new S3 bucket specifically for the static website.

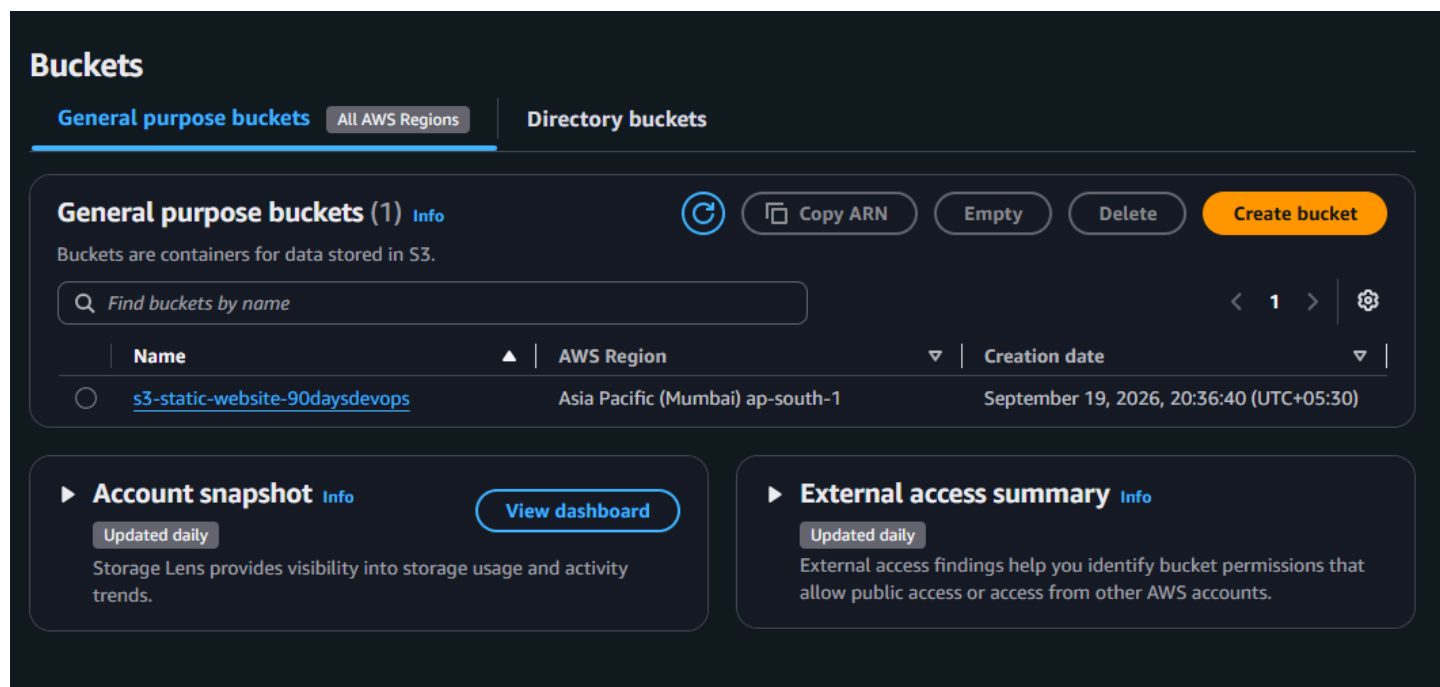
Configuration

Bucket name: s3-static-website-90daysdevops

AWS Region: Asia Pacific (Mumbai) — ap-south-1

For this learning exercise, public access configuration was adjusted because the S3 website endpoint needed to serve website objects publicly.

Screenshot



Newly created S3 bucket for hosting the static website.

6. Step 2 — Upload Website Files

The website files were uploaded into the S3 bucket.

Screenshot

☐	Name	Type	Last modified	Size	Storage class
☐	about.html	html	September 19, 2026, 20:41:03 (UTC+05:30)	8.7 KB	Standard
☐	blog.html	html	September 19, 2026, 20:41:02 (UTC+05:30)	10.6 KB	Standard
☐	contact.html	html	September 19, 2026, 20:41:02 (UTC+05:30)	7.7 KB	Standard
☐	css/	Folder	-	-	-
☐	images/	Folder	-	-	-
☐	includes/	Folder	-	-	-
☐	index.html	html	September 19, 2026, 20:41:03 (UTC+05:30)	30.6 KB	Standard
☐	services.html	html	September 19, 2026, 20:41:03 (UTC+05:30)	10.8 KB	Standard

Website Files Uploaded to S3

7. Step 3 — Enable Static Website Hosting

I navigated to: S3 → Bucket → Properties → Static website hosting

Static website hosting was enabled.

Configuration

Hosting type: Host a static website

Index document: index.html

The index document tells S3 which HTML file should be served as the main webpage.

Screenshot

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

Enabled

Hosting type

Bucket hosting

Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://s3-static-website-90daysdevops.s3-website.ap-south-1.amazonaws.com>

8. Step 4 — Configure Bucket Permissions

By default, S3 buckets are not publicly readable.

For this particular learning project, I configured the bucket to allow public read access to the website objects.

A bucket policy was added with the following permission:

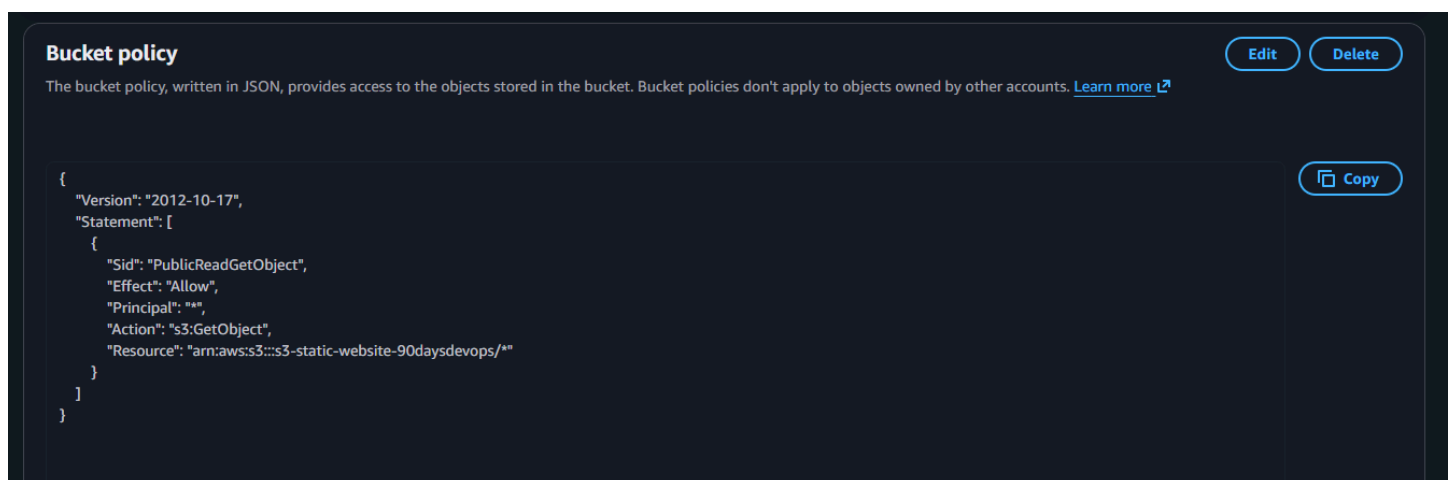
Action: s3:GetObject

The policy allows users to retrieve objects from the bucket.

Bucket Policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::s3-static-website-90daysdevops/*"
    }
  ]
}
```

Screenshot



Bucket policy allowing public s3:GetObject access to website objects.

9. Step 5 — Access the Website

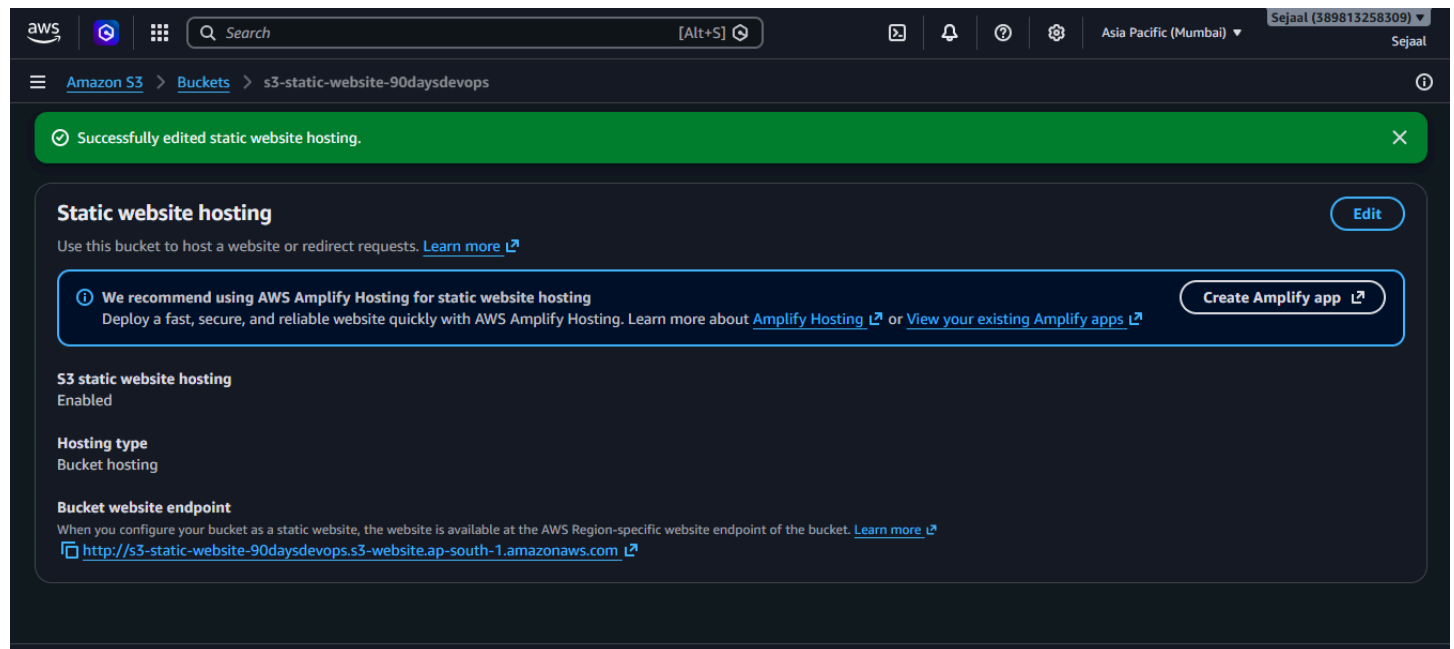
After configuring static website hosting and the required permissions, I accessed the website using the S3 website endpoint.

The endpoint was available under:

S3 → Properties → Static website hosting

The browser successfully loaded the index.html file and its associated CSS.

Screenshot

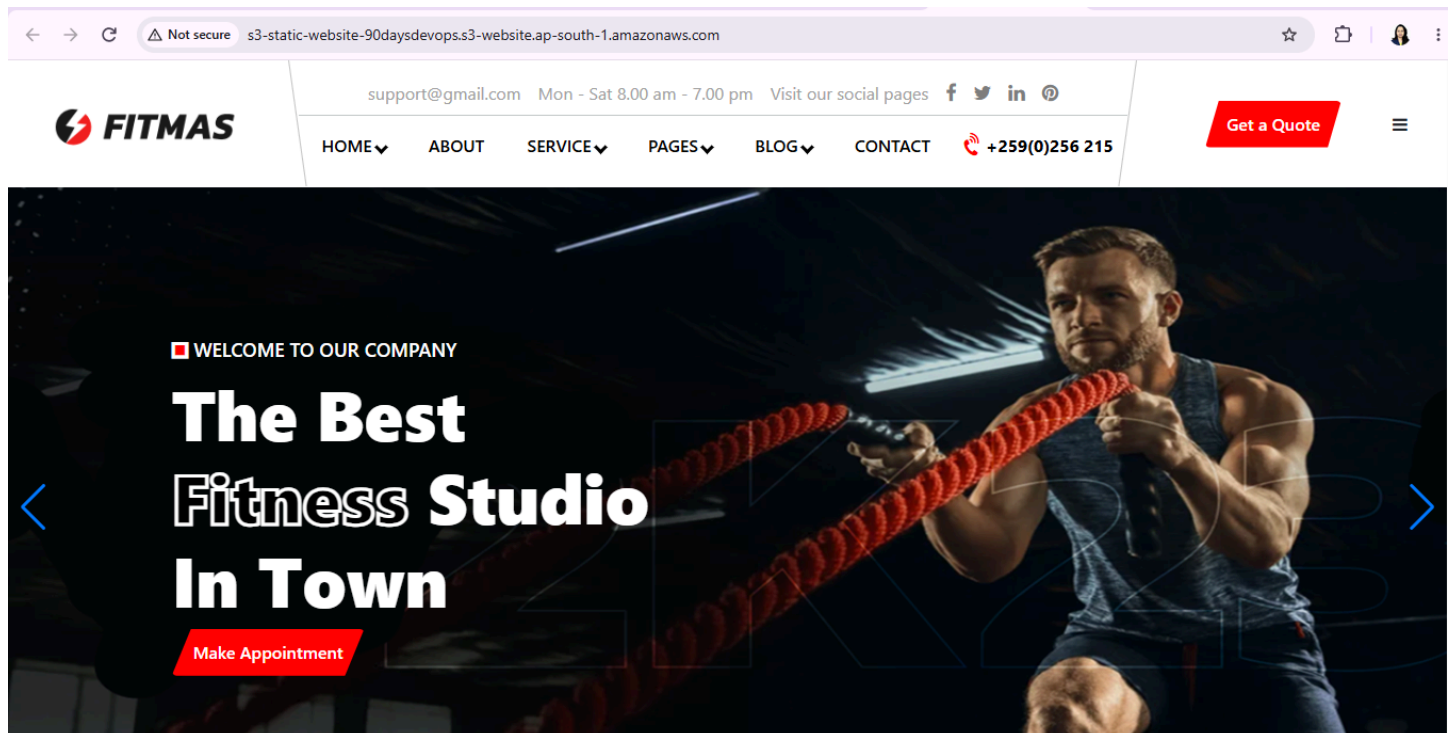


Static website successfully accessed through the S3 website endpoint.

10. Final Website Output

The final website displayed the content hosted directly from Amazon S3.

Screenshot



Final static website successfully served from Amazon S3.

11. Practical Verification

The following checks were performed during the project:

Verification	Result
S3 bucket created	✓ Successful
Website files uploaded	✓ Successful
index.html configured	✓ Successful
Static website hosting enabled	✓ Successful
Required access policy configured	✓ Successful
S3 website endpoint accessible	✓ Successful
HTML page rendered in browser	✓ Successful
CSS loaded correctly	✓ Successful

12. Troubleshooting — 403 AccessDenied

During the practical, the S3 website initially returned:

403 Forbidden

Code: AccessDenied

Message: Access Denied

Cause: The website files existed in the bucket and static website hosting was enabled, but the required permission for anonymous users to retrieve the objects had not yet been configured.

Resolution:

I configured the appropriate bucket policy allowing:

s3:GetObject

for the website objects.

After configuring the required permissions, the website endpoint successfully served the website.

What I Learned

This helped me understand that:

Enabling static website hosting alone does not automatically make S3 objects publicly readable.

Website hosting configuration and access permissions are separate concepts.

13. Key Concepts Learned

1. Amazon S3

Amazon S3 is an object storage service that can store files such as:

- HTML
- CSS
- JavaScript
- Images
- Documents
- Application assets

2. Static Website

A static website generally consists of files that are delivered to the browser without server-side application processing.

Examples include:

- HTML
- CSS

- JavaScript
- Images

3. Static Website Hosting

S3 provides a website hosting capability that can serve static content through an S3 website endpoint.

4. Index Document

The index document specifies the default webpage that S3 should return.

In this project: **index.html**

5. Bucket Policy

A bucket policy is a resource-based policy that defines permissions for accessing resources in an S3 bucket.

In this project, the policy allowed: **s3:GetObject** for the website objects.

6. S3 Website Endpoint

The website was accessed through the S3 static website endpoint rather than through an EC2 server.

14. What Makes This Different from EC2 Hosting?

One of the main takeaways from this practical was understanding the difference between traditional server hosting and S3 static website hosting.

Traditional EC2 Approach

User

↓

Internet

↓

EC2

↓

Apache / Nginx

↓

Website Files

You need to manage:

- EC2 instance
- Operating system
- Web server
- Security groups
- Server updates
- Storage

S3 Static Website Approach

User



S3 Website Endpoint



S3 Bucket



Website Files

There is no EC2 instance or traditional web server to manage.

15. Important Security Consideration

For this project, public access was intentionally configured to demonstrate the S3 website endpoint.

However, publicly accessible S3 buckets should not be used for sensitive or private data. For production applications, a common architecture is:

User



CloudFront



Private S3 Bucket

This approach can provide features such as:

- HTTPS
- CDN caching
- Custom domain integration
- Better control over access to the S3 origin

Therefore, the public S3 website configuration used in this project should be considered a learning/demo setup.

16. Production Improvement

A production-oriented version of this project could be extended with:

Amazon CloudFront

Use CloudFront in front of S3 for content delivery and HTTPS.

Custom Domain

Connect a custom domain using Amazon Route 53 or another DNS provider.

HTTPS

Serve the website securely using HTTPS.

CI/CD

Automate website deployment so that changes pushed to GitHub are automatically uploaded to S3.

Infrastructure as Code

Use Terraform or AWS CloudFormation to create and manage the infrastructure.

17. Project Outcome

The practical successfully demonstrated how a static website can be hosted using Amazon S3.

Final Result: Static website successfully hosted and accessed through Amazon S3. 

18. Key Takeaway

The biggest takeaway from this project was understanding that Amazon S3 is not limited to storing files.

For static websites, S3 can serve HTML, CSS, JavaScript, and other static assets without requiring an EC2 instance or traditional web server.

This practical also helped me understand the relationship between:

S3

|

├── Objects

├── Static Website Hosting

├── Index Document

└── Bucket Policies

19. Cleanup

After completing the practical, temporary AWS resources can be removed to avoid leaving an unnecessary publicly accessible bucket.

Cleanup steps

1. Delete website objects.
2. Empty the S3 bucket.
3. Delete the S3 bucket

Note: Cleanup is especially important for learning environments where the resources are no longer required.

20. Project Summary

Item	Details
Project	Hosting a Static Website on Amazon S3
AWS Service	Amazon S3
Region	ap-south-1
Website Type	Static
Main File	index.html
Styling	style.css
Hosting	S3 Static Website Hosting
Access	S3 Bucket Policy
Web Server	Not required
EC2	Not required
Status	Completed

21. Skills Demonstrated

AWS

- Amazon S3
- S3 Bucket Management
- S3 Static Website Hosting
- S3 Permissions
- Bucket Policies
- Object Management

Cloud Concepts

- Object Storage
- Static Website Hosting
- Public vs Private Access
- Resource-Based Policies
- Cloud-based application hosting

Practical Skills

- AWS Console
- Website deployment
- Troubleshooting
- Access configuration
- Cloud architecture understanding

22. Project Evidence

Screenshots Included

- 01 – S3 Bucket Created
- 02 – Website Files Uploaded
- 03 – Static Website Hosting Configuration
- 04 – S3 Bucket Policy
- 05 – S3 Website Endpoint
- 06 – Final Hosted Website

23. Conclusion

This hands-on project gave me practical experience with Amazon S3 static website hosting and helped me understand how object storage can be used to serve static web content.

More importantly, troubleshooting the initial 403 AccessDenied response helped reinforce the relationship between S3 website hosting configuration and access permissions.

This is a simple architecture, but it provides a strong foundation for understanding more production-oriented architectures involving S3, CloudFront, HTTPS, DNS, and CI/CD.

